Statistics in Psychology and Education

Course Syllabus

Instructor: Hau-Hung Yang

Course Information

Course Level: Undergraduate
Credits: 3 credits (one semester)
Prerequisites: None
Software: R (with RStudio)

Course Description

This course provides psychology and education students with a comprehensive introduction to applied statistical methods commonly used in psychological and educational research. The course emphasizes intuition, correct interpretation, and hands-on data analysis with R. Students will learn to formulate research questions in statistical terms, choose appropriate analytic strategies, conduct analyses, and critically evaluate statistical results reported in published research. By the end of the course, students will possess the practical skills to analyze data independently and the conceptual understanding to be informed consumers of quantitative research.

Course Objectives

Upon completion of this course, students will be able to:

1. Understand the logic of statistical inference, including estimation and hypothesis testing.
2. Select appropriate statistical methods for common research designs in psychology and education.
3. Conduct statistical analyses using R and interpret output correctly.
4. Critically evaluate the use of statistics in published research articles.
5. Communicate statistical results clearly in written reports.

Textbook and References

- Howell, D. C. (2013). *Statistical Methods for Psychology* (8th ed.). Cengage Learning. **[Primary textbook]**
- Field, A., Miles, J., & Field, Z. (2012). *Discovering Statistics Using R*. SAGE Publications. [Supplementary reference]

Assessment

Homework assignments (weekly R exercises)	30%
Midterm examination	25%
Final examination	25%
In-class quizzes and participation	10%
Data analysis project	10%

Weekly Schedule

Week	Topic
1	Introduction to statistics: Why do we need statistics? Populations, samples, and the logic of inference; getting started with R and RStudio
2	Descriptive statistics and data visualization: Central tendency and variability; histograms, boxplots, and scatterplots in R
3	Probability foundations: Basic probability rules, conditional probability, Bayes' theorem
4	Random variables and probability distributions: Discrete and continuous distributions; the normal distribution
5	Sampling distributions and the Central Limit Theorem
6	Point estimation and interval estimation: Confidence intervals for means
7	Introduction to hypothesis testing: Null and alternative hypotheses, Type I and Type II errors, p-values
8	*Midterm Examination*
9	One-sample and two-sample t-tests
10	Paired-samples t-test; statistical power and effect size; project question and dataset approval
11	One-way analysis of variance (ANOVA): Between-subjects designs

Week	Topic (continued)
12	Multiple comparisons and post-hoc tests
13	Correlation: Pearson and Spearman coefficients; project draft for feedback
14	Simple linear regression: Model, estimation, and interpretation; project AI-analysis critique
15	Chi-square tests; review of method selection and common misuses; final project due
16	*Final Examination*

The schedule follows the 16-week NTU semester: fourteen teaching weeks, the midterm examination in Week 8, and the final examination in Week 16. A class lost to a national holiday is made up in the flexible week after Week 16 (Week 16+1), as the NTU academic calendar allows.

Data Analysis Project (10%)

Using an instructor-approved public dataset, each student will address a psychological or educational question with a method covered in class. Submit reproducible R code and a 3–5 page report stating the question, method and assumptions, results, and limitations. Critique one statistical claim from a published study relevant to the dataset or question. Compare an AI-generated analysis with your own, identify unsupported reasoning, and provide a justified correction; instructor-provided AI output is available.

The question and dataset are approved in Week 10, a draft is due for feedback in Week 13, the AI critique is discussed in Week 14, and the final report and code are due in Week 15. The project grade assesses question and method choice (30%), reproducibility (30%), critical evaluation and correction (25%), and clear interpretation and writing (15%). These weights apply within the project's 10% course allocation.

Course Policies

- All homework and project analyses must be submitted as reproducible R scripts or R Markdown documents.
- Late submissions will incur a 10% penalty per day, up to a maximum of three days.
- Academic integrity is expected; all individual work must be the student's own.
- AI assistance must be disclosed and checked independently. Use public or simulated data for external AI tools; instructor-provided outputs can be used for the critique.